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S403 GALLERY WORKS

1. SCOPE OF WORK

- 1.1 The Bidder shall design, furnish and install complete system of gallery work for all buildings, structures, and equipment furnished by the Bidder including all accessories as indicated in Part 4, Subsection 1.2 “ Summary of Work “ and S-001 “General Design Requirement of Structure, Civil and Building Works” including but not be limited to, the following:
- 1.2 Framing for steel stairs, galleries and platforms.
- 1.3 Grating and grating treads, including abrasive nosings.
- 1.4 Safety floor plate (checkered plate).
- 1.5 Handrails.
- 1.6 Ladders.
- 1.7 Guard plates.
- 1.8 Banding of all openings in grating occasioned by field run piping.
- 1.9 All materials necessary, including Type 316 stainless steel studs for grating hold-down clips to connect the gallery work to structural steel and other parts of the structure already in place.
- 1.10 All bracket, hangers and bracing necessary to support the Work rigidly and keep it in free from vibration and excessive deflection.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 ‘Codes and Standards’.
- 2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 ‘Codes and Standards’ are specifically applicable to the design, manufacture and testing of the work included in this specifications:

ASTM		American Society for Testing and Materials
AISC		American Institute of Steel Construction Manual
AWS		American Welding Society
SSPC		Steel Structures Painting Council, USA
SAE		Society of Automotive Engineers, USA
UBC		Uniform Building Code of USA
OSHA		Occupational Safety and Health Administration, USA
JIS		Japanese Industrial Standards

- 2.3 Code requirements: All stair riser heights, stair tread depths, stair widths, ladders, ladder guards, hand railing heights etc. shall conform to the latest UBC and OSHA code requirements that may apply.

The gallery work shall also conform to the mandatory requirements of all Indonesian authorities having jurisdiction. It shall be Bidder's responsibility to investigate such requirements and determine the applicable provisions. These provisions shall be current to the date of contract award.

3. WORKMANSHIP, SEQUENCE AND FINISH

- 3.1 Workmanship and finish shall be equal to the best shop and field practice. All material shall have clean surfaces before being worked. Fabricated members shall be free from twists, bends or open joints. Shearing and clipping shall be neatly and accurately done. All work shall be neatly finished.
- 3.2 As much of the work as possible, including connections and joints between members, attachment of clip angles, punching of holes etc. shall be shop fabricated so as to reduce the amount of field erection work to a minimum.
- 3.3 The Bidder shall be responsible for proper fit of all materials furnished, including railing and post clearance with other equipment, piping, grating, checkered plate etc. including field check for clearances.
- 3.4 The Bidder shall fabricate and erect the gallery works in the sequence desired by the Owner if requested.

4. MATERIALS

- 4.1 Grating shall be galvanized steel unless otherwise specified. Steel grating shall be fabricate from welding quality steel equal to SAE-1015 for bearing bars and SAE-1010 for cross bars.
- 4.2 Structural steel and all other steel including floor plates, ladders, cages, guard plates, etc. shall conform to applicable requirements of ASTM A36 – Standard Specifications for Structural Steel.
- 4.3 Bolts shall be of ASTM A325 Type 1 steel.
- 4.4 Handrails and post shall be butt welded steel pipe of standards weight (Schedule 40) or steel structural tubing of equal (such as shock type connection) or greater weight unless galvanized pipe is called for in the specifications. The pipe shall conform to ASTM A120 with its steel meeting a minimum tensile strength of 3000 kg/cm² and a yield point of 1750 kg/cm² on actual physical tests. The structural tubing shall conform to ASTM A501, F-36 or better material. The Bidder shall retain final responsibility for spacing of handrails posts. All handrails shall be shop final responsibility for spacing of handrail posts. All handrails shall be shop prime and field finish painted.

5. DESIGN REQUIREMENTS

5.1 Grating

Grating, except where noted otherwise on the approved drawings, shall have a minimum thickness of 38 mm with 38 mm x 5 mm main bars on 30 mm center-to-center bar spacing and crossbars on not greater than 100 mm center-to-center spacing. Deflection of grating shall not exceed L/300. The proper size of grating shall be subjected to Design Loading.

5.2 Stairs

Channels stringers and grating tread stairs shall be used in all plant areas to service equipment or to travel from one gallery level to another..

This type of stair having a maximum slope of 40°, a maximum and a minimum riser of 190 mm and 165 mm respectively and a maximum and minimum tread of 250 mm and 225 mm respectively is considered gallery stairs. The stair treads shall be rigidly bolted to stringers with bolt heads on the

outside and nuts on the inside faces of stringers. The stair width shall not be less than 75 cm.

5.3 Handrails

5.3.1 Handrails shall be provided for the following areas and where required:

5.3.1.1 Around every stairway floor opening.

5.3.1.2 Around every floor opening into which personnel could accidentally trip or fall with an opening cover.

5.3.1.3 On all flights of stairs having four or more risers.

5.3.1.4 At all open sided floors, walkways and platforms.

5.3.1.5 Adjacent to dangerous equipment.

5.3.1.6 All temporary openings shall be guarded by temporary handrails.

5.3.2 Handrails shall be placed at 105 cm above the nosing of treads, or above the top of platform grating or checkered plate. They shall be continuous the full length of the stairs. Ends shall be returned or shall terminate in newel posts or safety terminals.

5.3.3 Handrails shall be standard pipe handrails 40 mm nominal diameter and 3.5mm of thickness, welded construction. The handgrip portion of handrails shall have a continuous smooth surface with no sharp corners.

5.3.4 Handrails projecting from a wall shall have a space of not less than 40 mm between the wall and the handrails.

5.4 Guard Plates

5.4.1 Guard plates shall be 6 mm thick and shall extend a minimum of 100 mm above the finished floor line, unless otherwise required.

5.4.2 Guard plates shall be fastened to steel framing members or concrete slabs.

- 5.4.3 All ground plates shall be attached to framing members in the shop, where possible, in order to minimize the amount of field erection work.

5.5 Ladders

- 5.5.1 Ladders shall be provided in all pits, manholes, and entrances to tunnels, to all roofs over 6m above the ground or required by the Owner where there is no access by a stair, or to the floors where no stair system is provided.
- 5.5.2 The following types of ladders shall be used in the design of the power plant:
- 5.5.2.1 Ladders which consist of individual round steel rungs, minimum size is 19 mm diameter.
- 5.5.2.2 Ladders which consist of steel side rails and steel rungs attached to side rails.
- 5.5.2.3 Ladders equipped with a cage.
- 5.5.3 Ladder service platforms or landings 3.6 m or higher above ground shall have a safety (cage). Ladders over 6 m in height which are equipped with a cage must have a rest platform at intervals not exceeding 10 m.

6. DETAILED REQUIREMENTS

6.1 Grating

- 6.1.1 Grating shall be a standard make of fabricated galvanized grating of a type acceptable to the Owners however grating of the pressure locked type is not acceptable. The welding requirements hereinafter specified do not apply to weld grating which shall be produced in accordance with standard manufacturing practice.
- 6.1.2 Grating panels shall be of uniform width as far as possible. Grating bars on grating areas adjacent to each other shall run in the same direction, if economically possible, and grating panels shall in other respects be so fabricated that the design will match when panels are assembled. Ends of panels shall be cut off neatly. Ends which have become bent must be straightened before acceptance. All grating shall be straight and not warped and shall have full bearing on the supporting steel.

- 6.1.3 In order to obtain a firm bearing on the supporting steel, each section of grating shall be fastened near each corner to the supporting steel. The method of fastening shall be by either tack welding, fasteners or a combination of both.
- 6.1.4 Fasteners shall be permanently fastened to the grating sections so as not to become detached and lost when removing and replacing grating section.
- 6.1.5 Special care shall be taken to anchor overhanging grating sections securely by means of fasteners. Grating shall be provided with stops to prevent any section from slipping off the supporting steel when an adjacent panel is removed.
- 6.1.6 All gallery steel connecting to structural steel shall be continuous fillet welded.
- 6.1.7 Grating sections shall fit snugly with not more than 3 mm clearance between adjacent sections of grating, and 8 mm clearance grating and curb angles or guard plates.
- 6.1.8 Holes and grating for pipes, where required by Bidder's design, shall be cut and banded in the shop, unless otherwise specified.
- 6.1.9 All other holes that are not specifically located and dimensioned on the Bidder's design drawings shall be cut and banded in the field in accordance with field measurements of pipes in place.
- 6.1.10 When laying out holes in grating for piping, consideration must be given to clearances required to accommodate pipe covering (if any) and movement of the piping due to thermal expansion.
- 6.1.11 Grating shall be split around all pipes where possible and shall be made into easily removable sections. Location of junction lines between grating sections shall be determined from actual field measurements of pipes insofar as it is possible. All holes for pipes shall clear the pipe (or pipe covering as the case may be) by not less than 50 mm or more than 75 mm. Furthermore, the arrangement shall allow for the displacement of pipes due to their expansion or contraction without contact between grating and pipe (or pipe covering).

- 6.1.12 All holes shall be banded with a 5 mm bar projecting 25 mm above top of grating. The grating bars shall be neatly welded to the band. All bands shall be divided on the junction line between adjacent panels.
- 6.1.13 When it is necessary in the installation of a floor grating panel to cut the grating at right angles to the bars at a point of support, suitable means of permanently holding the ends of the bar in alignment shall be provided.
- 6.1.14 **Nosing**
- 6.1.14.1 Edges of grating at head of stairs shall have a 40 mm wide and not less than 8 mm thick abrasive nosing.
- 6.1.14.2 Where grating junction lines occur at head of stairs, the nosing shall be cut at the junction line so as to permit removal of each section of grating.
- 6.1.14.3 Provide manufacturer's standard paint finish on metal areas of tread as nosing. Abrasive areas shall not be painted. Any discolorations, etc., on the treads and nosing shall be cleaned off with a solvent after erection, and before completion of the work.
- 6.1.14.4 Install stair tread nosing in the shop but only after the grating treads have been galvanized.
- 6.1.14.5 The projecting edges of longitudinal grating bars at openings not protected by guard plates shall be banded with 6 mm thick bar of same depth as the grating.
- 6.1.15 Outdoor grating and stair treads shall have serrated top. All exterior grating shall be clipped down with a minimum of two saddle clips per grating section.
- 6.2 Stairs and Treads**
- 6.2.1 Stairs shall be constructed and attached to galleries or concrete work as indicated on the Bidder's drawings approved by the Owner.
- 6.2.2 Stair treads shall be 32 mm x 5 mm standard galvanized grating of the same pattern as the floor grating. Grating treads shall have abrasive noising.
- 6.2.3 Stair treads shall be rigidly bolted to stringers with bolt heads on outside and nuts on inside faces of stringers.

6.3 Checkered Steel Floor Plates

- 6.3.1 Checkered steel floor plate shall be rolled steel galvanized plate of a type and design suitable for the loading. Steel floor plates shall have a nominal thickness of 6 mm unless otherwise required.
- 6.3.2 The pattern center lines on all checkered steel floor plate sections shall match and line up in both directions on adjacent sections.
- 6.3.3 All floor plate sections shall fit snugly into place with not more than 1.5 mm space between.
- 6.3.4 All steel floor plates shall be amply reinforced by means of steel angles along all edges of each plate and in intermediate locations where necessary. Each stiffener shall be welded to underside of plate along both edges of the stiffener. Each section shall be provided with two lifting holes located in opposite corners. Floor plates shall be straight and flush with the surrounding floor. Each floor plate section shall be provided with grinding bars, for aligning faces of plate sections which each other and with adjacent flooring. Grinding bars shall have bearing over their entire area.
- 6.3.5 Steel floor plates, where curb angle construction is required shall also be reinforced with stiffeners except for narrow trenches where stiffeners may not be required.
- 6.3.6 All openings for pipes shall be neatly cut so as to clear the pipe (or pipe covering) 12 mm all around. The arrangements shall be such that any section can be readily removed.

6.4 Handrails

- 6.4.1 All railings and railing post connections shall be in accordance with the details show on the approved drawings.
- 6.4.2 All railing and post intersections shall be welded. Pipe or tubing posts shall be welded to the top rail, the rail being continuous. The center rail shall be welded to the posts, the posts being continuously.
- 6.4.3 The railing and post sections shall be completely fabricated in the shop in as large sections as is practicable and that can be conveniently handled and erected. Field railing splices where required shall be welded and shall in all

cases be located near a pipe post or points of support. No screwed couplings will be permitted in rails or posts.

- 6.4.4 After shop and/or field welding is done, all rough weld metal projections shall be neatly ground off so that the joints will present a smooth appearance. Neat appearing welds with no rough projections will not require grinding. All sharp corners shall be rounded.
- 6.4.5 The Bidder shall cut handrails and guard plates and weld (except galvanized steel) in field to clear diagonal bracing where required.
- 6.4.6 Ends of handrails at building columns or other vertical steel may be attached to the steel by means of a collar welded to the steel.
- 6.4.7 Where gates are required, swing gates, shall be provided. No chain gate shall be used.

6.5 Guard Plates

- 6.5.1 Guard plates shall be furnished in as long length as possible. Ends of adjoining guard plates shall be welded together. Joints shall be no closer than 1 m to handrail post.
- 6.5.2 Guard plates used at edges of floors and galleries and meeting at angles at corners shall have corners welded. If necessary, an extra piece of plate shall be welded to the outside of guard plates to close the gap between plates that do not join.
- 6.5.3 Cutouts shall be provided where necessary to clear piping or other equipment.
- 6.5.4 The Bidder shall provide ground plates around all column penetrations and moment connections. Gratings shall be coped and fitted with cover plates so as to prevent openings greater than 25 mm in width around these penetrations.

6.6 Ladders

- 6.6.1 Unless otherwise specified, ladders shall be galvanized and made of steel bar stringers and round steel bar rungs.
- 6.6.2 Suitable guards or cages shall be furnished for ladders for locations and conditions stated in Clause 5.5.3 or where required by UBC. Guards shall be of type and design as required by these codes.
- 6.6.3 Access from platform, floors or landings to all ladders going to a lower elevation shall be provided with a pipe swing gate, as may be indicated or as may be required by OSHA or UBC.

7. FABRICATIONS AND ERECTION

Fabrications and erections shall be in accordance with the applicable provisions of the AISC Manual of Steel Construction (latest edition) and the following:

7.1 Shop Connections

- 7.1.1 All shop connections shall be made by welding or with bolts
- 7.1.2 Gallery members connecting to existing work or to steel work by others shall be welded. For welding requirements, see Clause 7.1.3.
All other gallery work shall be bolted. For bolted connection requirements, see Clause 7.1.4
- 7.1.3 Welded connections shall conform to Tables III and IV, “Framed Beam Connections”, of the AISC Manual, using E70XX electrodes unless otherwise indicated. The connection angle length shall be the maximum for each size beam.
- 7.1.4 Bolted connections shall conform to table IIA, “Framed Beam Connections”, of the AISC Manual, using double angle connections with minimum 19 mm diameter bolts of ASTM A325 Type 1 for friction –type connections unless otherwise indicated. Number of bolts and connection angle length shall be the maximum for each size beam. The minimum projection of the installed bolt beyond the nut shall be 6 mm.

7.2 Bearing Plates

Beams bearing on masonry or concrete work shall be provided with bearing plates and clip angles.

7.3 Fabrication

Structures shall be plumbed or leveled such that the deviations for the complete work do not exceed the requirements of the AISC Code of Standard Practice. Erection drawings shall be marked accordingly.

7.4 Welding

Welding shall be in accordance with the applicable provisions of the AISC Manual, AWS D1.1, "Structural Welding Code" and Specification S-410.

8. CLEANING AND PAINTING

8.1 Shop cleaning and painting of gallery steel shall be in accordance with the Technical Specification S-420.

8.2 The field finish coat of paint for surface other than grating, ladder stringers and bar rungs shall be applied per Technical Specification S-430.

9. GALVANIZING

9.1 Grating, stair tread, ladders with cage, floor plates and stop bars shall be complete shop fabricate and galvanized after fabrication unless noted otherwise. The galvanizing shall be compatible to the environmental conditions.

9.2 All parts to be galvanized shall be designed for erecting with bolts, pins, and anchor bolts, clips etc. and shall no require field welding, field riveting, field drilling or any operation that would damage galvanizing.

9.3 All bolts, clips, nuts washers, anchor bolts, stops, banding and other appurtenance required for galvanized parts shall also be galvanized or stainless steel and shall be furnished and installed by Bidder.

9.4 Galvanizing shall conform to the applicable requirements set forth in the latest editions of the ASTM Standard Specifications A123, A153, A384, and A385.

- 9.5 Bolts shall be galvanized after threading. Nuts shall be galvanized before threading. Members which are being punched or drilled shall be galvanized after punching or drilling.
- 9.6 Any grating which is warped or bent after galvanizing will be rejected and shall not be used.
- 9.7 Galvanized grating shall be tested in accordance with the requirements set forth in the latest edition of the ASTM Standard Specification A239.
- 9.8 The Bidder shall require to re-galvanize all gallery work that has had galvanizing damaged. If it becomes necessary to alter any galvanized stair treads or grating in the field, such work shall be re-galvanized before installing at no cost to the Owner.
- 10. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)**
- 10.1 Quality assurance/quality control shall conform to the requirements of Part 4 Subsection 2.7.
- 10.2 The quality control procedures and documentation shall include but not be limited to the following:
- 10.2.1 Mill certificate of the material
- 10.2.2 Welding.
- 10.2.3 Nondestructive testing.
- 10.2.4 Packaging and shipping.
- 10.2.5 Shop test procedures.
- 10.2.6 Cleaning, painting and galvanizing.

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